

System and Network Administration CoSc3212_

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System and Network Administration

- designed to receive process, manage and present information in a meaningful format.
- Hardware: the physical, tangible computer equipment and devices
 - Functions:
 - input,
 - processing (internal storage, computation and control),
 - output,
 - secondary storage (for data and programs), and
 - communication.

System and Network Administration

- designed to receive process, manage and present information in a meaningful format.
- Hardware: the physical, tangible computer equipment and devices
 - Components:
 - Input,
 - Processing,
 - Storage,
 - Output and
 - Communication devices.

System and Network Administration

- Software, also known as programs or applications, are the intangible components of the computer system.
 - Classification:
 - system software and
 - application software.

System and Network Administration

- Network: two or more computers connected together in such a way that they can share resources
 - Computer network: a collection of computers or other hardware devices that are connected together, either physically or logically, using special hardware and software, to allow them to exchange information and cooperate
- Networking: processes involved in:
 - designing,
 - implementing,
 - upgrading,
 - managing and
 - working with networks and network technologies

System and Network Administration

- Network:

- Resources:

- a file,
 - a folder,
 - a printer,
 - a disk drive, or
 - just about anything else that exists on a computer

System and Network Administration

- Network:
 - Classification of networks
 - by Geographical Span:
 - Personal Area Network (PAN): spanned across a given table
 - distances between the devices not more than few meters.
 - i.e. Bluetooth
 - belongs to a single person.
 - Local Area Network (LAN):
 - span across a building, or
 - across several buildings within a single organization,
 - using intermediate devices, like switches and/or hubs, to interconnect devices in all floors.

System and Network Administration

- Network:
 - Classification of networks
 - by Geographical Span:
 - Metropolitan Area Network (MAN): span across a whole city interconnecting several buildings and organizations.
 - Wide Area Network (WAN): span across multiple cities, or an entire country, or an entire continent or it may even cover the whole world.
 - Internet is one example of WAN.

System and Network Administration

- Network:
 - Classification of networks
 - by Inter-connectivity (topology):
 - Bus Topology: all devices are connected to a central cable, called the bus or backbone, which is terminated at its ends.
 - Message transmitted along the Bus is visible to all computers connected to the backbone cable.
 - the workstation checks the destination address contained in the message either to process or drop the packet if it matches or not respectively.

System and Network Administration

- Network:
 - Classification of networks
 - by Inter-connectivity (topology):
 - Bus Topology:
 - Advantages:
 - ease of installation
 - less amount of cable requirement.
 - Drawback
 - entire network will be shut down if there is a break in the main cable.

System and Network Administration

- Network:
 - Classification of networks
 - by Inter-connectivity (topology):
 - Star Topology: each node is connected directly to a central device called a hub or a switch.
 - Data passes through the central device (switch) before continuing to its destination.
 - The central device manages and controls all functions of the network.
 - common with twisted pair cable.
 - RJ-45 Connectors are used to connect the cable to the Network Interface Card (NIC) of each computer.

System and Network Administration

- Network:
 - Classification of networks
 - by Inter-connectivity (topology):
 - Star Topology:
 - Advantages:
 - ease of installation and reconfiguration,
 - robust (ease of fault identification and isolation),
 - link failure only affects device(s) connected to that link,
 - less expensive than mesh.
 - Drawbacks:
 - more cable requirements (than bus and ring)
 - single point of failure (if central device fail, the whole system will be down).

System and Network Administration

- Network:
 - Classification of networks
 - by Inter-connectivity (topology):
 - Ring Topology: all devices are connected to one another in the shape of a closed loop,
 - each device is connected directly to two other devices, one on either side of it.

System and Network Administration

- Network:
 - Classification of networks
 - by Inter-connectivity (topology):
 - Ring Topology:
 - Advantages
 - easy to install and reconfigure,
 - less expensive (than mesh),
 - performance is even despite the number of users.
 - Drawbacks:
 - break in the ring (such as a disabled station) can disable the entire network,
 - limitations on media and traffic (limitation on ring length and number of devices).

System and Network Administration

- Network:
 - Classification of networks
 - by Inter-connectivity (topology):
 - Mesh Topology: devices connected with many redundant interconnections between network nodes.
 - Full mesh: every node has a connection to every other node
 - most expensive
 - number of cables grows fast as the number of nodes increases,
 - $((n(n - 1)) / 2)$, , where n is the number of nodes

System and Network Administration

- Network:
 - Classification of networks
 - by Inter-connectivity (topology):
 - Mesh Topology:
 - Advantages:
 - dedicated links between devices,
 - robust (single link failure don't affect entire network),
 - privacy/security (direct communication between communicating devices),
 - ease of fault identification and isolation.
 - Drawbacks:
 - installation and reconnection are difficult (large number of cables),
 - huge amount of cables consumes a lot of space, and
 - it is the most expensive of all.

System and Network Administration

- Network:
 - Classification of networks
 - by Inter-connectivity (topology):
 - Hybrid Topology: contains more than one topology
 - inherits merits and demerits of all the incorporating topologies.

System and Network Administration

- Network:
 - Classification of networks
 - by Administration:
 - Private network: belongs to a single autonomous system and
 - cannot access outside of its physical or logical domain.
 - Public network: can be accessed by anyone inside or outside of an organization

System and Network Administration

- Network:
 - Classification of networks
 - by Architecture:
 - Client-Server Architecture:
 - one or more systems acting as Server.
 - Other being Client, request the Server to serve requests.
 - Servers take and process request on clients' behalf.
 - Peer-to-Peer (Point-to-point): Two systems can be connected Point-to-Point (back-to-back).
 - both reside on same level and called peers.
 - can be hybrid network which involves network architecture of both the above types

System and Network Administration

- Network:
 - Internet Control Message Protocol (ICMP):
 - management protocol and messaging service provider for IP
 - works at the Network layer and is used by IP for many different services

System and Network Administration

- Network:
 - Internet Control Message Protocol (ICMP):
 - events and messages:
 - Destination Unreachable: router can't send an IP datagram any further,
 - it uses ICMP to send a message back to the sender
 - Buffer Full: router's memory buffer for receiving incoming datagrams is full,
 - it will use ICMP to send out this message until the congestion abates.

System and Network Administration

- Network:
 - Internet Control Message Protocol (ICMP):
 - events and messages:
 - Hops: Each IP datagram is allotted a certain number of routers, called hops, to pass through.
 - If it reaches its limit of hops before arriving at its destination, the last router to receive that datagram deletes it.
 - The executioner router then uses ICMP to send an obituary message, informing the sending machine of the demise of its datagram.

System and Network Administration

- Network:
 - Internet Control Message Protocol (ICMP):
 - events and messages:
 - Ping (Packet Internet Groper): uses ICMP echo messages to check the physical and logical connectivity of machines on a network.
 - Traceroute Using ICMP timeouts, is used to discover the path a packet takes as it traverses an Internetwork.

System and Network Administration

- Network:
 - Process/Application Layer Protocols:
 - Telnet
 - File Transfer Protocol (FTP)
 - Network File System (NFS).
 - Simple Mail Transfer Protocol (SMTP)
 - POP3 is used to receive mail.
 - Simple Network Management Protocol (SNMP)
 - Domain Name Service (DNS)
 - Dynamic Host Configuration Protocol (DHCP)

System and Network Administration

- **System Administration**

- Philosophy

- Network and system administration is a branch of engineering that concerns the operational management of human-computer systems
 - about putting together a network of computers, getting them running and then keeping them running in spite of the activities of users who tend to cause the systems to fail
 - A system administrator: works for users, so that they can use the system to produce work
 - about hardware, software, user support, diagnosis, repair and prevention.
 - need to know a bit of everything
 - skills are technical, administrative and socio psychological

System and Network Administration

- **System Administration**

- Philosophy

- System vs Network Administration:

- System administration: term used traditionally by mainframe and Unix engineers to describe the management of computers whether they are coupled by a network or not
 - Network administration: management of network infrastructure devices (routers and switches)

System and Network Administration

- **System Administration**

- Philosophy

- The themes that make up the philosophy of system administration:

- Automate everything
 - Document everything
 - Communicate as much as possible
 - Know your resources
 - Know your users
 - Know your business
 - Security cannot be an afterthought
 - Plan ahead
 - Expect the unexpected
 - Backup and disaster recovery planning
 - Patching

System and Network Administration

- **System Administration**

- Scope of Network administration:
 - Monitoring the network capacity to ensure that all transmission requirements can be met.
 - Adding capacity to the network by
 - increasing band width
 - interconnecting additional nodes, or
 - creating and interconnecting additional networks.
 - Training people to use the network effectively

System and Network Administration

- **System Administration**

- Scope of Network administration:
 - Assisting IT professionals in organizational applications that will make good use of the network's capabilities
 - Backing up the network software and data regularly to protect against failure of network
 - Putting security procedures
 - Making sure the network personnel can respond quickly and effectively in the event of a network operational or security failure
 - Diagnosing and troubleshooting problems on the network

System and Network Administration

- **System Administration**

- Goal of Network administration
 - to keep the network running properly and
 - configuring and managing services that are provided over the network

System and Network Administration

- **System Administration**

- Challenges of system administration
 - Designing a network which is logical and efficient.
 - Deploying large numbers of machines which can be easily upgraded later.
 - Deciding what services are needed.
 - Planning and implementing adequate security.
 - Providing a comfortable environment for users.
 - Developing ways of fixing errors and problems which occur.
 - Keeping track of and understanding how to use the enormous amount of knowledge which increases every year.

System and Network Administration

- **System Administration**

- Meta principles of system administration
 - Principle 1 (Policy is the foundation)
 - System administration begins with a policy - a decision about
 - what we want and what should be, in relation to what we can afford.
 - Policy speaks of:
 - what we wish to accomplish with the system, and
 - what we are willing to tolerate of behavior within it

System and Network Administration

- **System Administration**

- Meta principles of system administration
 - Principle 2 (Predictability)
 - highest level aim in system administration is to work towards a predictable system
 - what makes system administration difficult is that it involves a kind of 'search' problem. It is the hunt for a stable region in the landscape of all policies
 - In choosing policy, one might easily promote a regime of cascading failure, of increasing unpredictability that degenerates into chaos
 - Avoiding these regimes is what makes system administration difficult

System and Network Administration

- **System Administration**

- Meta principles of system administration
 - Principle 3 (Scalability)
 - scalable systems are those that grow in accordance with policy;
 - i.e. they continue to function predictably, even as they increase in size
 - When we speak of predictability, it must always be within a margin of error.

System and Network Administration

- Windows Network Concepts
 - Workgroup
 - In networking:
 - a collection of computers on a LAN that share common resources and responsibilities
 - According to Microsoft:
 - peer-to-peer Windows workgroups in homes, schools and small businesses that may share files, printers, or Internet connection

System and Network Administration

- Windows Network Concepts
 - Windows workgroup: group of standalone computers in a peer-to-peer network.
 - joining a workgroup requires all participants to use a matching name
 - Windows 7, 8 and 10: WORKGROUP
 - Windows XP: MSHOME
 - To access resource:
 - know the name of the workgroup
 - the username and password of an account on the remote computer

System and Network Administration

- Windows Network Concepts
 - Windows workgroup: group of standalone computers in a peer-to-peer network.
 - Disadvantages of workgroups:
 - user account will need to be created on those machines with resources (same username and password).
 - low security protocol used for authentication between nodes.
 - Desktop computers have a fixed limit of 15 or less connections.

System and Network Administration

- Windows Network Concepts
 - Domain Controller: support client-server local networks.
 - Specially configured computer (Domain Controller) running a Windows Server operating system serves as a central server for all clients.
 - can handle more computers than workgroups
 - A client PC can belong to either to a workgroup or to a Windows domain, but not both.

System and Network Administration

- Windows Network Concepts
 - Microsoft HomeGroup: introduced in windows 7
 - simplify the management of workgroups for administrators particularly homeowners
 - HomeGroup security settings can be managed through one shared login instead of manually setting up shared user accounts
 - Joining a Homegroup does not remove a PC from its Windows WORKGROUP
 - PCs older than Windows 7 (like XP and vista), cannot be members of HomeGroups

System and Network Administration

- Windows Network Concepts
 - LDAP & Windows Active Directory
 - Lightweight Directory Access Protocol (LDAP)
 - open, vendor-neutral, industry standard application protocol for accessing and maintaining distributed directory information services over an Internet Protocol (IP) network
 - sharing of information about users, systems, networks, services, and applications throughout the network

System and Network Administration

- Windows Network Concepts
 - LDAP & Windows Active Directory
 - Lightweight Directory Access Protocol (LDAP)
 - Protocol:
 1. A client starts an LDAP session by connecting to an LDAP server (Directory System Agent - DSA)

System and Network Administration

- Windows Network Concepts
 - LDAP & Windows Active Directory
 - Lightweight Directory Access Protocol (LDAP)
 - Protocol:
 2. client sends an operation request to the server,
 - a) StartTLS: use TLS for a secure connection
 - b) Bind - authenticate and specify LDAP protocol version
 - c) Search - search for and/or retrieve directory entries
 - d) Compare - test if a named entry contains a given attribute value
 - e) Add a new entry

System and Network Administration

- Windows Network Concepts
 - LDAP & Windows Active Directory
 - Lightweight Directory Access Protocol (LDAP)
 - Protocol:
 2. client sends an operation request to the server,
 - f) Delete an entry
 - g) Modify an entry
 - h) Modify Distinguished Name (DN) - move or rename an entry
 - i) Abandon - abort a previous request
 - j) Extended Operation - generic operation used to define other operations
 - k) Unbind - close the connection (not the inverse of Bind)

System and Network Administration

- Windows Network Concepts
 - LDAP & Windows Active Directory
 - Lightweight Directory Access Protocol (LDAP)
 - Protocol:
 3. server sends responses
 - client does not need to wait for a response before sending the next request
 - server may send the responses in any order
 - information is transmitted using Basic Encoding Rules (BER)
 - common alternative method of securing LDAP communication is using an SSL tunnel.

System and Network Administration

- Windows Network Concepts
 - LDAP & Windows Active Directory
 - Windows Active Directory
 - Active Directory (AD): developed by Microsoft for Windows domain networks.
 - AD Domain Controller:
 - authenticates and authorizes all users and computers in a Windows domain type network,
 - assigning and enforcing security policies for all computers, and
 - installing or updating software.

System and Network Administration

- Windows Network Concepts
 - LDAP & Windows Active Directory
 - Windows Active Directory
 - Active Directory (AD): developed by Microsoft for Windows domain networks.
 - Services:
 - Domain Services (DS) - best known
 - Lightweight Directory Services (LDS)
 - Certificate Services (CS)
 - Federation Services (FS)
 - Rights Management Services (RMS)

System and Network Administration

- Windows Network Concepts
 - LDAP & Windows Active Directory
 - Windows Active Directory
 - Active Directory (AD):
 - Logical Structure: consists of a database and corresponding executable code responsible for servicing requests

System and Network Administration

- Windows Network Concepts
 - LDAP & Windows Active Directory
 - Windows Active Directory
 - Active Directory (AD):
 - Logical Structure:
 - Objects
 - resources (e.g., printers) and
 - security principals (user or computer accounts and groups)

System and Network Administration

- Windows Network Concepts
 - LDAP & Windows Active Directory
 - Windows Active Directory
 - Active Directory (AD):
 - Logical Structure:
 - Forests, trees, and domains: logical divisions
 - Domain: groups objects
 - Tree: collection of one or more domains in a contiguous namespace,
 - Forest: collection of trees that share a common global catalog, directory schema, logical structure, and directory configuration
 - represents the security boundary within which users, computers, groups, and other objects are accessible

System and Network Administration

- Windows Network Concepts
 - LDAP & Windows Active Directory
 - Windows Active Directory
 - Active Directory (AD):
 - Logical Structure:
 - Organizational Units: objects held within a domain can be grouped into organizational units
 - provides:
 - hierarchy to a domain,
 - ease its administration, and
 - can resemble the organization's structure in managerial or geographical terms
 - Microsoft recommends using OUs rather than domains for structure and to simplify the implementation of policies and administration.

System and Network Administration

- Windows Network Concepts
 - LDAP & Windows Active Directory
 - Windows Active Directory
 - Active Directory (AD):
 - Physical Structure: defined by one or more IP subnets.
 - Replication: synchronizes changes using multi-master replication
 - replicas pull changes from the server where the change was effected
 - Knowledge Consistency Checker (KCC) creates a replication topology of site.

System and Network Administration

- Windows Network Concepts
 - LDAP & Windows Active Directory
 - Windows Active Directory
 - Active Directory (AD):
 - AD Implementation
 - in general has more than one licensed Windows server computer.
 - Certain Microsoft products such as SQL Server and Exchange can interfere with the operation of a domain controller, necessitating isolation of these products on additional Windows servers.
 - Physical hardware costs for the many separate servers can be reduced through the use of virtualization

System and Network Administration

- Windows Network Concepts
 - LDAP & Windows Active Directory
 - Windows Active Directory
 - Active Directory (AD):
 - AD Trusting: to allow users in one domain to access resources in another
 - Trusts inside a forest are automatically created when domains are created

System and Network Administration

- Windows Network Concepts
 - LDAP & Windows Active Directory
 - Windows Active Directory
 - Active Directory (AD):
 - AD Trusting:
 - Terminology
 - One-way trust: One domain allows access to users on another domain, but the other domain does not allow access to users on the first domain.
 - Two-way trust: allow access to users on both domains.
 - Trusted domain: The domain that is trusted; whose users have access to the trusting domain.
 - Transitive trust: A trust that can extend beyond two domains to other trusted domains in the forest.
 - Intransitive trust: A one way trust that does not extend beyond two domains.

System and Network Administration

- Windows Network Concepts
 - LDAP & Windows Active Directory
 - Windows Active Directory
 - Active Directory (AD):
 - AD Trusting:
 - Terminology
 - Explicit trust: A trust that an admin creates. It is not transitive and is one way only.
 - Cross-link trust: explicit trust between domains in different trees or in the same tree when a descendant/ancestor (child/parent) relationship does not exist between the two domains.
 - Shortcut: Joins two domains in different trees, transitive, one- or two-way.
 - Forest trust: Applies to the entire forest. Transitive, one- or two-way.

System and Network Administration

- Windows Network Concepts
 - LDAP & Windows Active Directory
 - Windows Active Directory
 - Active Directory (AD):
 - AD Trusting:
 - Terminology
 - Realm: Can be transitive or nontransitive (intransitive), one- or two-way.
 - External: Connect to other forests or non-AD domains. Nontransitive, one- or two-way.
 - PAM trust: A one-way trust used by Microsoft Identity Manager from a (possibly low-level) production forest to a (Windows Server 2016 functionality level) 'bastion' forest, which issues time-limited group memberships.

System and Network Administration

- Windows Network Concepts
 - LDAP & Windows Active Directory
 - Windows Active Directory
 - Active Directory (AD):
 - AD Management solutions
 - Active Directory Administrative Center ($\geq$ Windows Server 2012),
 - Active Directory Users and Computers,
 - Active Directory Domains and Trusts,
 - Active Directory Sites and Services,
 - ADSI Edit,
 - Local Users and Groups,
 - Active Directory Schema snap-ins for Microsoft Management Console (MMC),
 - SysInternals ADEplorer

System and Network Administration

- **User Administration**

- User account: a collection of settings and information
 - which files and folders you can access,
 - what you can do on your computer,
 - what are your preferences, and
 - what network resources you can access when connected to a network.
- allows you to authenticate to Windows or any other operating system so that you are granted authorization to use them

System and Network Administration

- **User Administration**

- User account:

- Attributes:

- User name: the name you are giving to that account.
 - Password: the password associated with the user account (can also use blank passwords in ≤ Windows 7).
 - User group: a collection of user accounts that share the same security rights and permissions.
 - A user account must be a member of at least one user group.

System and Network Administration

- **User Administration**

- User account:

- Attributes:

- Type: defines their permissions and what they can do in Windows.

- Administrator: has complete control over the PC.

- can install anything and make changes that affect all users of that PC.

- Standard:

- can only use the software that's already installed by the administrator and

- change system settings that don't affect other users.

System and Network Administration

- **User Administration**

- User account:

- Attributes:

- Type: defines their permissions and what they can do in Windows.

- Guest: special type of user account that has the name Guest and no password.

- can only use the software that's already installed by the administrator and

- cannot make any changes to system settings.

System and Network Administration

- **User Administration**

- Multi-user operating systems such as Windows don't allow a user to use them without having a user account.

System and Network Administration

- **User Administration**

- File & Folder Permissions

- Permissions: for assigning access rights to specific user accounts and user groups.
 - defines which user accounts and user groups can access which files and folders, and what they can do with them
 - to learn the permissions of any folder, right click on it and select -Properties.
 - Properties → Security tab → Group or user names section → see all the user accounts and use groups that have permissions to that folder.

System and Network Administration

- **User Administration**

- File & Folder Permissions

- User account or group can receive one of the permissions:

- Read: allows the viewing and listing of a file or folder (and subfolders).
 - Write: allows writing to a file or adding files and subfolders to a folder.
 - List folder contents: this permission can be assigned only to folders. It permits the viewing and listing of files and subfolders, as well as executing files that are found in that folder.

System and Network Administration

- **User Administration**

- File & Folder Permissions

- User account or group can receive one of the permissions:

- Read & execute:

- File: permits the reading and accessing of a file's contents as well as its execution.

- Folders: allows the viewing and listing of files and subfolders, as well as the execution of files.

- Modify:

- Files, allows their reading, writing and deletion.

- Folders, allows the reading and writing of files and subfolders, plus the deletion of the folder.

- Full control

System and Network Administration

- **User Administration**

- Assigning User Rights

- efficient way: make the user a member of a group that already has the right.
 - or give the user the rights directly
 - approach used with the Remote Desktop Users group
 - assign user rights through the Local Policies node of Group Policy.

System and Network Administration

- **User Administration**

- Roaming profile

- set of files that contains all settings of a user including per-user configuration files and registry settings.
 - In an Active Directory or NT4 domain you can set that the profile of a user is stored on a server.
 - enables the user to log on to different Windows domain members and use the same settings
 - when a user logs in: a copy of the profile is downloaded from the server to the Windows domain member when a user logs into.
 - Until the user logs out, all settings are stored and updated in the local copy.
 - During the log out, the profile is uploaded to the server.

System and Network Administration

- **User Administration**

- Roaming profile
 - Application data and settings
 - Directories you create under c:\users\[your username] i.e. Documents, Downloads... have a roaming profile, the master copy of the profile is stored on a fileserver.
 - When you log in to a Windows computer, the contents of your profile will be synchronized from the fileserver to the local computer.
 - When you log out of the computer, any changes to the profile are then synchronized back to the server.
 - the contents of the profile are stored centrally, whenever you log on to any computer in the department you will have the same application data and settings (e.g., internet browser bookmarks, preferences in Microsoft Office etc.)
 - all data stored within File server is automatically backed up

System and Network Administration

- **User Administration**

- Roaming profile
 - Roaming vs local profiles
 - Main advantage to using a local profile is that the profile is accessible even when the computer is disconnected from the network.
 - A major drawback of a local profile is that the user profile data is not being automatically backed up by the server.

System and Network Administration

• **User Administration**

- Roaming profile
 - Roaming vs local profiles
 - If you have a large roaming profile, the login and logoff times may take a significant amount of time.
 - Solution:
 - Folder redirection with roaming profiles
 - allows specific folders (such as the Desktop and Documents folder) to be permanently stored on the server.
 - eliminates the need for the redirected folder to be copied as a part of the logon and logoff processes

System and Network Administration

• **User Administration**

- Windows Registry: hierarchical database that stores low-level settings for Windows operating system and for applications that opt to use the registry.
 - The kernel, device drivers, services, Security Accounts Manager, and user interface can all use the registry.
- Registry Structure
 - Keys and values
 - keys are container objects similar to folders
 - referenced with a syntax similar to Windows' path names,
 - using backslashes to indicate levels of hierarchy
 - may contain values and subkeys
 - values are non-container objects similar to files.

System and Network Administration

- **User Administration**

- Routing and NAT

- Network Address Translation (NAT): translates the IP addresses of computers in a local network to a single IP address
 - This address is often used by the router that connects the computers to the Internet.
 - Computers on the internet only see the IP address of the router.
 - adds an extra level of security, since the router can be configured as a firewall

System and Network Administration

- **User Administration**

- Routing and NAT

- Proxies and Gateways

- Proxy server: acts as a gateway between you and the Internet

- Used for:

- Firewalls

- Content filters

- Bypassing content filters

- Caching

- Security

- Sharing Internet connections

System and Network Administration

- **User Administration**

- Routing and NAT

- Proxies and Gateways

- Gateway: a node (router), a key stopping point for data on its way to or from other networks.

- Possible forms and tasks:

- Web application firewalls

- Cloud storage gateways

- IoT gateways

- Email security gateways

- VoIP trunk gateways

System and Network Administration

- **Resource Monitoring and Management**

- a great deal of system administration revolves around resources and their efficient use.
- System Performance Monitoring:
 - normally done in response to a performance problem
 - Either the system is running too slowly, or
 - programs (and sometimes even the entire system) fail to run at all.

System and Network Administration

- **Resource Monitoring and Management**

- System Performance Monitoring:

- Monitoring

- to identify the nature and scope of the resource shortages that are causing the performance problems

- Monitoring System Capacity:

- part of an ongoing capacity planning program
 - Capacity planning uses long- term resource monitoring to determine rates of change in the utilization of system resources.

- Once these rates of change are known it becomes possible to conduct more accurate long- term planning regarding the procurement of additional resources

System and Network Administration

- **Resource Monitoring and Management**

- System Performance Monitoring:

- Monitoring

- What to Monitor:

- CPU power, bandwidth, memory, and storage
 - How much free space is available?
 - How many I/O operations on average does it perform each second?
 - How long on average does it take each I/O operation to be completed?
 - How many of those I/O operations are reads? How many are writes?
 - What is the average amount of data read/written with each I/O?

System and Network Administration

- **Resource Monitoring and Management**

- System Performance Monitoring:

- Monitoring

- Monitoring CPU Power

- determine if CPU utilization reaches 100%, if not - can process more work

- Monitoring Bandwidth: more difficult than the other resources to monitor

- Statistics:

- Bytes received/sent
 - Interface counts and rates
 - Transfers per Second

System and Network Administration

- **Resource Monitoring and Management**

- System Performance Monitoring:

- Monitoring

- Monitoring Memory: a wealth of performance statistics can be found

- Statistics:

- Page Ins/Page Outs

- Active/Inactive Pages

- Free, Shared, Buffered, and Cached Pages

- Swap Ins/Swap Outs

System and Network Administration

- **Resource Monitoring and Management**

- System Performance Monitoring:

- Monitoring

- Monitoring Storage

- Levels: possible to have dire problems in one area and no problems whatsoever in the other

- Monitoring for sufficient disk space

- Monitoring for storage-related performance problems

- Statistics:

- Free Space

- File System-Related Statistics

- Transfers per Second

- Reads/Writes per Second

System and Network Administration

- **Resource Monitoring and Management**

- System Performance Monitoring:

- Monitoring Tools

- As your organization grows, so does the number of servers, devices, and services you depend on.
 - Users will notice performance problems with the software that they use, but those problems rarely arise within that software.
 - Knowing whether a computer has issues is fairly straightforward when the computer is right in front of you.
 - Even the smallest small-office/home-office network has multiple nodes: laptops, desktops, tablets, WiFi access points, Internet gateway, smartphones, file servers and/or media servers, printers, and so on. That means you are in charge of -infrastructure|| rather than just -equipment.|| Any component might start misbehaving and could cause issues for the others.

System and Network Administration

- **Resource Monitoring and Management**

- System Performance Monitoring:

- Monitoring Tools

- 1. Bandwidth monitoring tools and NetFlow and sFlow

- traffic analyzers help you stay aware of the activity, capacity, and health of your network

System and Network Administration

- **Resource Monitoring and Management**

- System Performance Monitoring:

- Monitoring Tools

- 2. Windows Task Manager: information about computer performance and shows detailed information about the programs and processes running on the computer

- name of running processes,
 - CPU load,
 - commit charge,
 - I/O details,
 - logged-in users,
 - Windows services,
 - network status and how the network works

System and Network Administration

- **Resource Monitoring and Management**

- System Performance Monitoring:

- Monitoring Tools

- 3. Windows Resource Monitoring: look at the presence and allocation of resources on a computer.

System and Network Administration

- **Resource Monitoring and Management**

- System Performance Monitoring:

- Monitoring Tools

- 3. Windows Resource Monitoring:

- Monitors:

- CPU Usage

- Memory

- Commit: Amount of virtual memory reserved by the operating system for the process.

- Working Set: Amount of physical memory currently in use by the process.

- Shareable: Amount of physical memory in use by the process that can be shared with other processes.

- Private: Amount of physical memory in use by the process that cannot be used by other processes.

System and Network Administration

- **Resource Monitoring and Management**

- System Performance Monitoring:

- Monitoring Tools

- 3. Windows Resource Monitoring:

- Monitors:

- Network Activity

- Bandwidth: maximum data transfer rate of a network

- most common Internet bottleneck is your connection to your Internet Service Provider

- Internet speed can also be different from the Internet bandwidth expectations

- Network bandwidth: fixed, can also be monitored by a network bandwidth monitor

- can also optimize data so that it consumes less bandwidth from what is allocated

- Network Printers

System and Network Administration

- **Resource Monitoring and Management**

- System Performance Monitoring:

- Remote Administration: control either a computer system or a network or an application or all three from a remote location.
 - adopted when it is difficult or impractical to a person to be physically present and do administration on a system's terminal

System and Network Administration

- **Resource Monitoring and Management**

- System Performance Monitoring:

- Remote Administration:

- Achieved using:

- SSH (Secure Shell)
 - OpenSSH (OpenBSD Secure Shell)
 - Telnet
 - rlogin
 - rsh
 - VNC (Virtual Network Computing)
 - FreeNX
 - Wireless Remote Administration
 - Remote Desktop Connection (RDC)

System and Network Administration

- **Resource Monitoring and Management**

- System Performance Monitoring:

- Redundant Array of Inexpensive (or Independent) Disks (RAID)
 - combines multiple physical disk drive components into one or more logical units for the purposes of data redundancy, performance improvement, or both.
 - in contrast with previous concept of highly reliable mainframe disk drives referred to as Single Large Expensive Disk (SLED)
 - Data is distributed across the drives in one of several ways, referred to as RAID levels
 - Originally, there were five standard levels of RAID, but many variations have evolved, including several nested levels and many non-standard levels (mostly proprietary).
 - standardised by: Storage Networking Industry Association (SNIA)
 - standard: RAID Disk Drive Format (DDF) standard

System and Network Administration

- **Resource Monitoring and Management**

- System Performance Monitoring:

- RAID 0:

- consists of striping, but no mirroring or parity.
 - sum of the capacities of the drives in the set.
 - striping distributes the contents of each file among all drives in the set,
 - failure of any drive causes the entire RAID 0 volume and all files to be lost.
 - throughput of read and write operations to any file is multiplied by the number of
 - reads and writes are done concurrently.

System and Network Administration

- **Resource Monitoring and Management**

- System Performance Monitoring:

- RAID 1

- consists of data mirroring, without parity or striping .
 - Data is written identically to two or more drives,
 - produces a mirrored set of drives.
 - any read request can be serviced by any drive in the set.
 - it can be serviced by the drive that accesses the data first (depending on its seek time and rotational latency),
 - Actual read throughput of most RAID 1 implementations is slower than the fastest drive.
 - Write throughput is always slower because every drive must be updated, and the slowest drive limits the write performance.
 - The array continues to operate as long as at least one drive is functioning

System and Network Administration

- **Resource Monitoring and Management**

- System Performance Monitoring:

- RAID 2 consists of bit-level striping with dedicated Hamming-code parity. All disk spindle rotation is synchronized and data is striped such that each sequential bit is on a different drive. Hamming-code parity is calculated across corresponding bits and stored on at least one parity drive. This level is of historical significance only; as of 2014 it is not used by any commercially available system.

System and Network Administration

- **Resource Monitoring and Management**

- System Performance Monitoring:

- RAID 3

- consists of byte-level striping with dedicated parity.
 - All disk spindle rotation is synchronized and data is striped such that each sequential byte is on a different drive.
 - Although implementations exist, RAID 3 is not commonly used in practice.

System and Network Administration

- **Resource Monitoring and Management**

- System Performance Monitoring:

- RAID 4

- consists of block-level striping with dedicated parity.
 - advantage of RAID 4 over RAID 2 and 3 is I/O parallelism:
 - in RAID 2 and 3, a single read I/O operation requires reading the whole group of data drives,
 - while in RAID 4 one I/O read operation does not have to spread across all data drives.
 - more I/O operations can be executed in parallel, improving the performance of small transfers.

System and Network Administration

• **Resource Monitoring and Management**

- System Performance Monitoring:

- RAID 5

- consists of block-level striping with distributed parity.
 - Unlike RAID 4, parity information is distributed among the drives, requiring all drives but one to be present to operate.
 - Upon failure of a single drive, subsequent reads can be calculated from the distributed parity such that no data is lost.
 - requires at least three disks.
 - large RAID 5 implementations are susceptible to system failures
 - array rebuild time and the chance of drive failure during rebuild.
 - Rebuilding an array requires reading all data from all disks, opening a chance for a second drive failure and the loss of the entire array.

System and Network Administration

• **Resource Monitoring and Management**

- System Performance Monitoring:

- RAID 6

- consists of block-level striping with double distributed parity.
 - Double parity provides fault tolerance up to two failed drives.
 - This makes larger RAID groups more practical, especially for high-availability systems,
 - requires a minimum of four disks.
 - The larger the drive capacities and the larger the array size, the more important it becomes to choose RAID 6 instead of RAID 5.
 - RAID 10 also minimizes these problems.

System and Network Administration

- **Security**

- Unix OS: a set of programs that act as a link between the computer and the user.
- Kernel: allocates the system resources and coordinates all the details of the computer's internals
- Shell: users use it to communicate with the kernel

System and Network Administration

- **Security**

- Linux distribution: operating system that is made up of a collection of software based on Linux kernel
 - contains the Linux kernel and supporting libraries and software

System and Network Administration

- **Security**

- Linux distribution:
 - Architecture

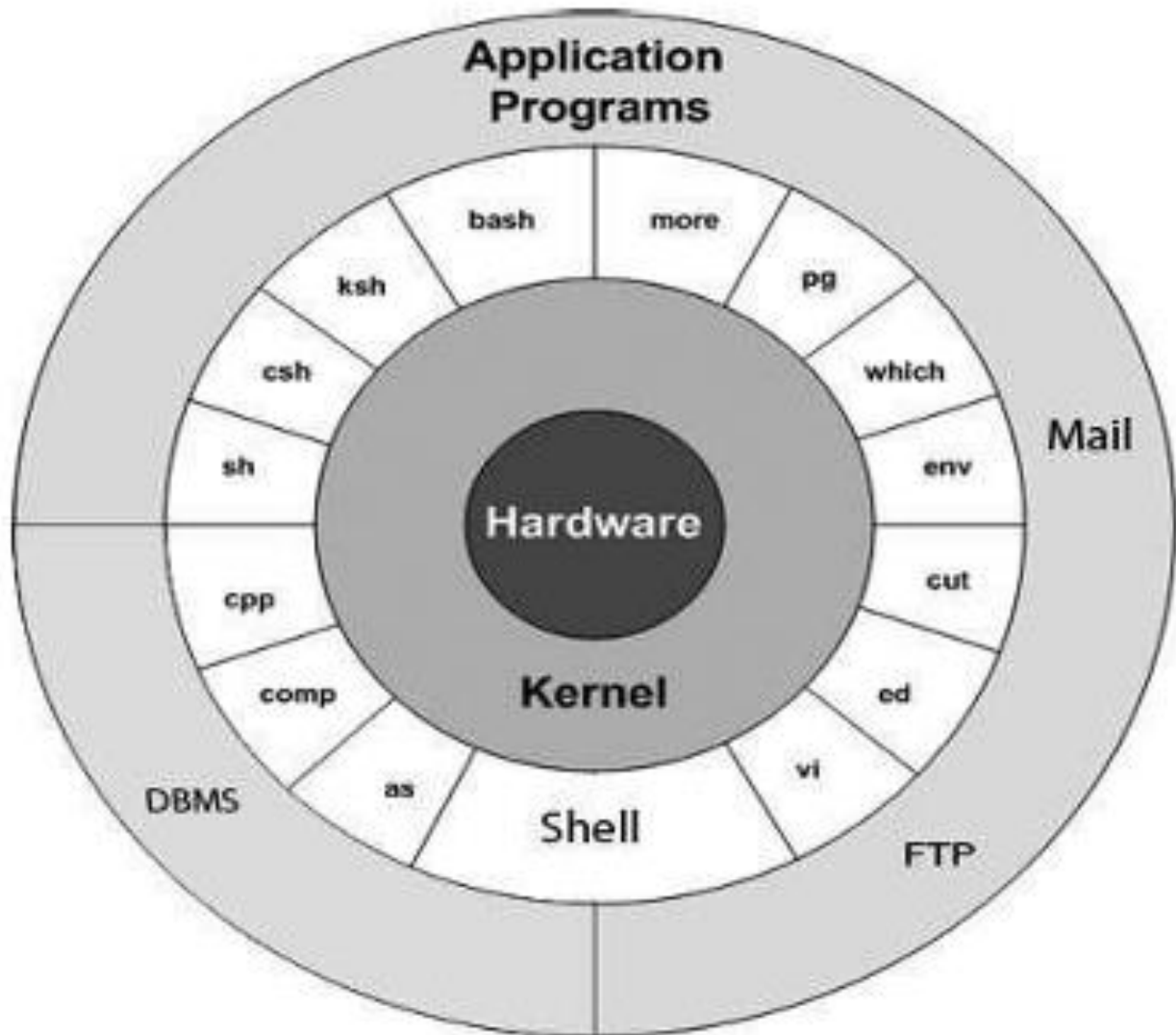


Figure. Block diagram of Unix system

System and Network Administration

- **Security**

- Linux distribution:

- Properties of Linux

- Linux Pros

- A lot of the advantages are due to its root in UNIX:

- Linux is free

- Linux is portable to any hardware platform

- Linux was made to keep on running

- Linux is secure and versatile

- Linux is scalable

- very short debug-times

- Linux Cons

- far too many different distributions

- not very user friendly and confusing for beginners

System and Network Administration

- **Security**

- Linux distribution:
 - Files

Symbol	Meaning
-	Regular file
d	Directory
l	Link
c	Special file
s	Socket
p	Named pipe
b	Block device

System and Network Administration

- **Security**

- Linux distribution:

- Directory Structure

- /bin:

- common programs,

- shared by the system, the system administrator and the users.

- /boot:

- startup files and the kernel, vmlinuz.

- /dev:

- references to all the CPU peripheral hardware represented as files

- /etc:

- most important system configuration files

- data similar to those in the Control Panel in Windows.

System and Network Administration

- **Security**

- Linux distribution:

- Directory Structure

- /home: Home directories of the common users.
 - /lib:
 - Library files,
 - includes files for all kinds of programs needed by the system and the users.
 - /lost+found:
 - every partition has a lost+found
 - files that were saved during failures are here.
 - /misc: for miscellaneous purposes.
 - /mnt:
 - standard mount point for external file systems,
 - e.g. a CD-ROM or a digital camera.

System and Network Administration

- **Security**

- Linux distribution:

- Directory Structure

- /opt: typically contains extra and third party software.
 - /proc: virtual file system containing system resources information.
 - /root: the administrative user's home directory.
 - /sbin: programs for use by the system and the system administrator.

System and Network Administration

- **Security**

- Linux distribution:

- Directory Structure

- /tmp :

- temporary space to be used by the system,
 - its contents will be cleaned upon reboot,
 - don't use this for saving any work!

- /usr: Programs, libraries, documentation etc. for all user-related programs.

- /var:

- storage for all variable and temporary files created by users,
 - log files,
 - temporary files downloaded from the Internet, or
 - to keep an image of a CD before burning it.